

Top 10 Mathematical Expressions Used in ML/DL

🔑 Core formulas.

💡 Widely used.

🎯 Know what each term means!

#	Expression & Terms (What each term denotes)	Used In / Purpose ⚙️	📖 Origin / Notes
1	Linear Regression (Prediction) $\hat{y} = w^T x + b$ $\hat{y}$: Predicted value (scalar) w : Weight vector ($n \times 1$) x : Feature vector ($n \times 1$) b : Bias (scalar)	Used in: - Linear Regression - Ridge / Lasso Regression Purpose: Predict a continuous target as a linear combination of features.	Origin: Linear models in statistics (Gauss, 1809 - Method of Least Squares) Foundation of most prediction models. Forms the building block for many other models.
2	Logistic Regression (Probability) $P(y = 1 x) = \sigma(w^T x + b) = \frac{1}{1 + e^{-(w^T x + b)}}$ $P(y=1 x)$: Probability of class 1 $\sigma(z)$: Sigmoid function w, x, b : As defined above	Used in: - Logistic Regression - Binary Classification Purpose: Map any real-valued number to a probability between 0 and 1.	Origin: Statistics / GLM (Developed in 1930s-1950s) Models probability of binary outcomes using a sigmoid transformation.
3	Cost Function (MSE) $J(w, b) = \frac{1}{n} \sum_{i=1}^n (\hat{y}^{(i)} - y^{(i)})^2$ J : Cost (mean squared error) n : Number of training examples $\hat{y}^{(i)}$: Predicted value for example i $y^{(i)}$: True value for example i	Used in: - Linear Regression (Also used in DL for regression) Purpose: Measure the average squared difference between prediction and actual value.	Origin: Least Squares Method (Gauss, 1809) One of the most common loss functions for regression problems.
4	Cross-Entropy Loss (Binary) $J = -\frac{1}{n} \sum_{i=1}^n [y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})]$ J : Loss $y^{(i)}$: True label (0 or 1) $\hat{y}^{(i)}$: Predicted probability of class 1 n : Number of examples	Used in: - Logistic Regression - Binary Classification (DL) Purpose: Measure the difference between true distribution and predicted probability.	Origin: Information Theory (Shannon, 1948) Standard loss for classification. Penalizes confident wrong predictions heavily.
5	Softmax Function (Multi-class Probability) $P(y = k x) = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}$ $P(y = k x)$: Probability of class k z_k : Logit for class k K : Total number of classes	Used in: - Multi-class Classification (DL) - Output layer of neural networks Purpose: Convert logits (real numbers) into probabilities that sum to 1.	Origin: Statistics / Multinomial Logistic Regression Generalization of sigmoid to multi-class problems.
6	Backpropagation (Gradient Update) $w \leftarrow w - \eta \frac{\partial J}{\partial w}$ w : Weights η : Learning rate J : Loss function $\frac{\partial J}{\partial w}$: Gradient of loss w.r.t. weights	Used in: - Training Neural Networks - Deep Learning Purpose: Update weights in the direction that reduces the loss.	Origin: Calculus (Chain Rule) Widely used since 1980s Backprop computes these gradients efficiently using the chain rule.
7	ReLU Activation Function $f(x) = \max(0, x)$ $f(x)$: Output x : Input to the neuron	Used in: - Hidden layers of Neural Networks - CNNs, Transformers, etc. Purpose: Introduce non-linearity while avoiding vanishing gradients.	Origin: Nair & Hinton, 2010 Simple and effective activation function. Most widely used.
8	Adam Optimizer Update Rule $\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \\ w_t &= w_{t-1} - \eta \frac{m_t}{\sqrt{\hat{v}_t} + \epsilon} \end{aligned}$ g_t : Gradient at time t m_t : First moment (mean) v_t : Second moment (variance) $\hat{m}_t, \hat{v}_t$: Bias-corrected moments η : Learning rate, ϵ : Small constant	Used in: - Training deep neural networks - Almost all modern DL models Purpose: Adaptive learning rate using estimates of first and second moments.	Origin: Kingma & Ba, 2014 Combines ideas from Momentum and RMSProp. Fast and robust optimizer.
9	Bayes' Theorem $P(H D) = \frac{P(D H) P(H)}{P(D)}$ $P(H D)$: Posterior (after data D) $P(D H)$: Likelihood $P(H)$: Prior $P(D)$: Evidence (normalizing const.)	Used in: - Bayesian Machine Learning - Naïve Bayes, Bayesian Nets Purpose: Update belief (H) after observing data (D).	Origin: Thomas Bayes, 1763 Foundation of Bayesian inference and probabilistic reasoning.
10	Transformer Attention (Scaled Dot-Product) $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$ Q : Query matrix K : Key matrix V : Value matrix d_k : Dimension of keys	Used in: - Transformers (NLP, LLMs) - Vision Transformers, etc. Purpose: Compute relevance of all tokens to each token (self-attention).	Origin: Vaswani et al., 2017 ("Attention Is All You Need") Core of modern LLMs (GPT, BERT, T5, LLaMA, etc.).

These 10 expressions are the building blocks of modern ML & DL. Understand the terms, their purpose, and where they come from!

